

Moshe Sofer

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Summary

B.Sc. Electrical & Electronics Engineering student at Tel Aviv University and member of the "Atidim" Excellence Program. Hands-on experience in hardware validation and high-speed signal analysis. Fast learner with a proven ability to ramp up quickly in new technical domains. Looking for a student position in hardware and chip development.

Education

2022-Present | B.Sc. Electrical and Electronics Engineering | Tel Aviv University.

Expected Graduation: 2027-2028 | Current GPA: 81

Relevant Courses: Digital Electronic Circuits, Digital Logic Systems, Data Structures and Algorithms, Analog Electronic Circuits, Introduction to VLSI Design (schematic design, layout, DRC/LVS), Computer Organization (MIPS, pipelining, cache & virtual memory).

Projects:

FPGA-Based Stopwatch System (Verilog, RTL): Implemented synthesizable RTL modules including adders, counters, FSM control, and debouncing; performed functional verification using self-checking testbenches and waveform analysis.

VLSI Design 8-bit ALU: Designed a transistor-level ALU featuring a high-speed Kogge-Stone Adder. Performed full custom layout design, including DRC/LVS verification and timing analysis (Logical Effort, Elmore Delay) to meet performance constraints.

Technical Skills

Programming: Python, C.

EDA & Design Tools: MATLAB, Cadence Virtuoso, Xilinx Vivado.

Hardware & Lab Tools: Logic Analyzer, Oscilloscope, power and signal measurement instruments, PCB testing and debugging, schematic and layout analysis (Altium Designer).

Work Experience

2025-2026 | Hardware Intern | MSolutions | Ra'anana, Israel.

- Participated in validation and debugging of USB3, HDMI, and HDBaseT communication boards based on Valens ASICs.
- Performed hands-on signal integrity, DC-DC power validation, and timing analysis using oscilloscopes and advanced lab instruments.
- Analyzed schematics and PCB layouts using Altium Designer, supporting board-level design understanding.
- Collaborated with hardware and firmware teams on board bring-up, system integration, and resolution of high-speed signal and design issues.
- Gained exposure to Valens ASIC architecture and ASIC-to-board integration, contributing to improved board reliability.

Military service

2016-2019 | Combat Commander (Sergeant Major) | "Duvdevan" Elite Unit | IDF | Israel.

Led teams in the elite Duvdevan unit, operating in structured, high-risk environments requiring precision, discipline, and accountability. Gained experience in execution under constraints, situational analysis, and responsibility for people and outcomes. Received a Certificate of Excellence and formal commendation for outstanding service and leadership.

2023-2025 | Reserve Duty, "Iron Swords" | IDF | Israel.

Maintained full-time Electrical Engineering studies alongside reserve duty.

Additional Information

Soft Skills: Analytical thinking, Fast learner, Ability to work under pressure, Leadership experience.

Volunteer: Coordination and support activities for individuals on the autism spectrum.

Languages: Hebrew - Native, English - Full Professional Proficiency.